ADVANCED NAVIGATION



MOTUS
MEMS IMU

Motus is a miniature ultra high accuracy MEMS IMU. It features some of the highest accuracy MEMS accelerometers and gyroscopes currently available combined with magnetometers.

Motus is fully calibrated for all sensor errors over a wide temperature range. It is available in both OEM and enclosed packages.

ITAR FREE

OEM AVAILABLE

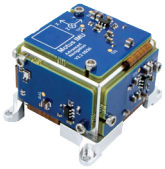
PERFORMANCE

- 0.05 ° Roll and Pitch
- 0.8 ° Heading (magnetic)
- 0.2 °/hr MEMS Gyroscope
- 1000 Hz Update Rate
- 150 g 6 ms Shock Limit

KEY FEATURES

- IMU, AHRS and INS options
- Rugged and OEM versions
- 500 ms Hot Start

OEM AVAILABLE



APPLICATIONS

AIR
- UAV Geopointing
- UAV Lidar
- Gimbal Stabilisation

LAND
- Gimbal Stabilisation
- Structural Monitoring
- Vehicle Navigation

SEA
- AUV Navigation
- ROV Navigation
- Hydrography

SPECIFICATIONS

NAVIGATION

Roll & Pitch Accuracy	0.05°
Heading Accuracy (magnetic)	0.8 °
Roll & Pitch Accuracy (external GNSS aided)	0.03 °
Heading Accuracy (external GNSS aided)	0.05 °
Horizontal Position Accuracy (external GNSS aided)	0.8 m
Vertical Position Accuracy (external GNSS aided)	1.5 m
Velocity Accuracy (external GNSS aided)	0.007 m/s
Orientation Range	Unlimited
Hot Start Time	500 ms
Internal Filter Rate	1000 Hz
Output Data Rate	Up to 1000 Hz

HARDWARE

Operating Voltage (OEM)	4.5 to 5.5 V
Operating Voltage (Enclosed)	5 to 36 V
Input Protection (Rugged only)	± 60 V
Power Consumption (Typical)	1.4 W
Hot Start Standby Duration	48 hrs
Operating Temperature	-40 °C to 85 °C
Environmental Protection (Enclosed)	IP67 MIL-STD-810G
MTBF	200,000 hrs
Shock Limit	150 g, 6 ms, half sine
Vibration Limit	8 g RMS
Dimensions (OEM)	34 x 39 x 24 mm
Dimensions (Enclosed)	42 x 67 x 30 mm
Weight (OEM)	26 grams
Weight (Enclosed)	98 grams

COMMUNICATION

Interface (OEM)	UART
Interface (Enclosed)	RS232 (RS422 version available)
Speed	4800 to 2M baud
Protocol	AN Packet Protocol or NMEA
Peripheral Interface	2x GPIO and Auxiliary RS232
GPIO Level	5 to 20 V
GPIO Functions	1PPS Input Sensor sync input Sensor sync output Odometer Stationary Air Data Input NMEA input/output Novatel GNSS input Trimble GNSS input AN Packet Protocol

SENSORS

SENSOR	ACCELEROMETERS	GYROSCOPES	MAGNETOMETERS
Range	± 10 g	± 475 °/s	± 8 G
Bias Instability	8 ug	0.2 °/hr	-
Initial Bias	< 0.45 mg	< 3 °/hr	-
Initial Scaling Error	< 0.03 %	< 0.02 %	< 0.07 %
Scale Factor Stability	< 0.04 %	< 0.03 %	< 0.09 %
Non-linearity	< 0.05 %	< 0.03 %	< 0.08 %
Cross-axis Alignment Error	< 0.05 °	< 0.05 °	< 0.05 °
Noise Density	2 ug/√Hz	6 °/hr/√Hz	210 uG/√Hz
Bandwidth	250 Hz	200 Hz	110 Hz